

ASR Benchmark for Indian Conversational Speech

Approach

I recorded 20 Bangalore locality utterances in natural Hinglish sentences (e.g. “Haan bhai, main Koramangala mein rehta hoon”), deliberately varying conditions — quiet room, fan/AC hum, street noise, rushed and whispered delivery, phone-speaker playback. These are the primary evaluation set and already carry real-world noise, so they are run as-is. I additionally test on FLEURS Hindi (30 samples) as an out-of-domain check, with 50% of samples augmented with Gaussian noise (seed-fixed) for a controlled clean-vs-noisy comparison.

Model selection

Deepgram Nova-2 is the mandated baseline (hosted API). I added three models chosen to isolate distinct variables rather than to pile on options:

- **Whisper Medium** (OpenAI, open-source) — the obvious self-hosted default; tests what a general multilingual model gives for free.
- **IndicWhisper** (vasista22/whisper-hindi-large-v2) — a Hindi fine-tuned Whisper. Isolates whether Indic fine-tuning actually buys accuracy over vanilla Whisper, and at what compute cost.
- **Sarvam AI Saaras v3** (Indic-specific hosted API) — tests whether an India-focused commercial vendor beats the global incumbent on exactly the accent/code-switching this platform sees.

Metrics

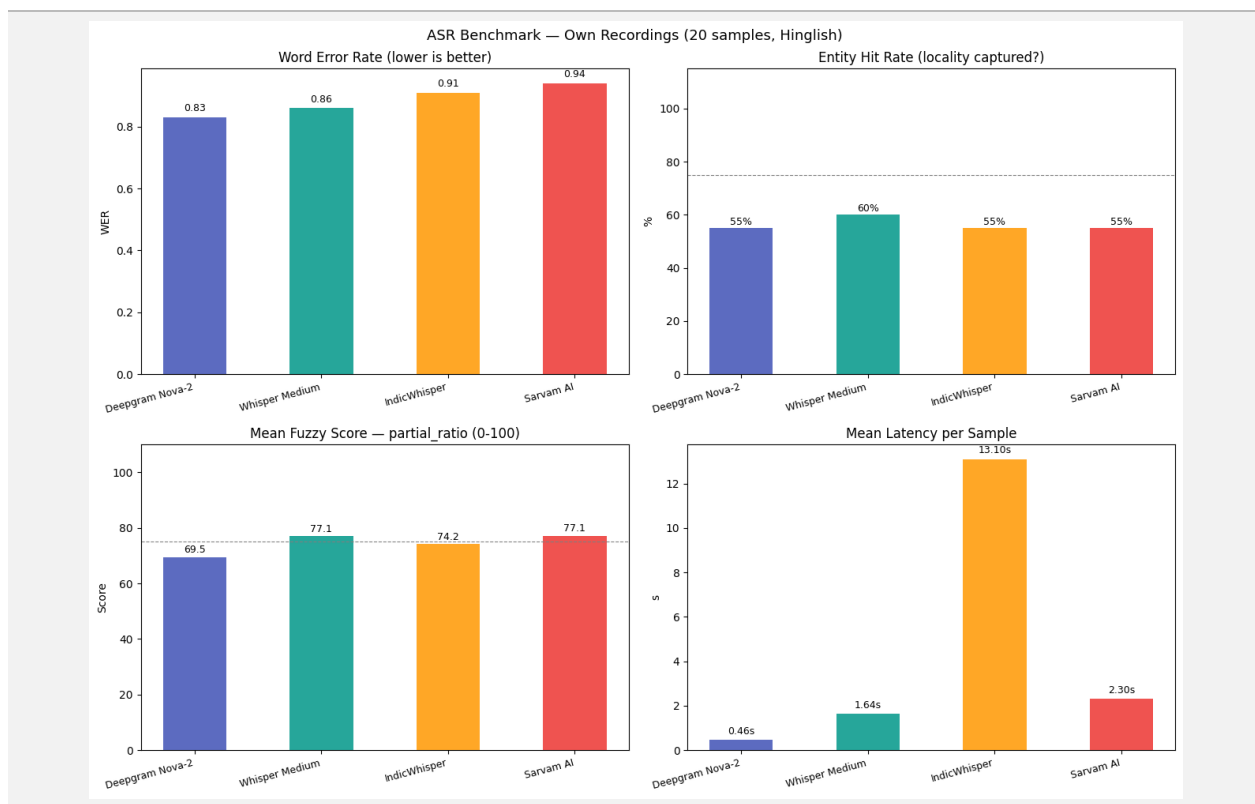
WER alone is misleading here. A model can mangle filler words yet still nail the locality — or score a deceptively low WER while dropping the one word that matters. So alongside **WER** and **CER**, I report two entity-focused metrics:

- **Entity Hit Rate (EHR)** — binary: was the locality name captured? This is the production-relevant number.
- **Fuzzy Score** — rapidfuzz partial_ratio (0–100) of locality against the hypothesis, threshold 75 for a “hit”. Captures near-misses that exact match would discard.

Normalization caveat (important): models return mixed Devanagari/Roman script, but ground truth is Roman Hinglish. Scoring raw output inflates WER toward 1.0 even for correct transcriptions. I normalize every hypothesis to Roman via word-level ITRANS transliteration before scoring.

Results

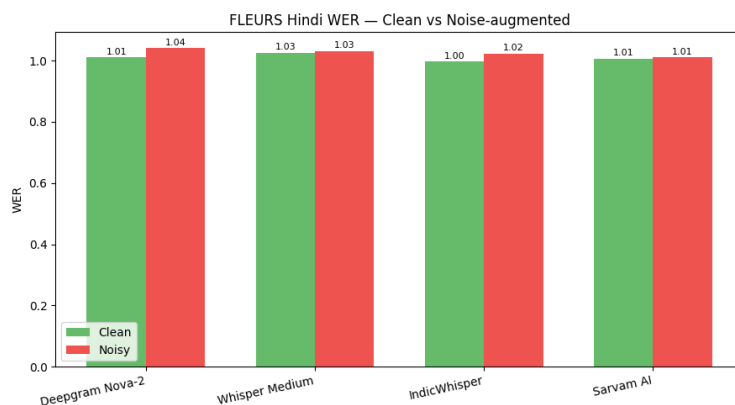
Model	WER	CER	Fuzzy	EHR	Latency
Deepgram Nova-2 (baseline)	0.831	0.399	69.5	55%	0.46s
Whisper Medium	0.861	0.410	77.1	60%	1.64s
IndicWhisper (hi large-v2)	0.910	0.776	74.2	55%	13.10s
Sarvam AI (Saaras v3)	0.940	0.417	77.1	55%	2.30s



Deepgram wins on speed and transcription; Whisper Medium wins on entity capture. Deepgram has the lowest WER (0.831) and crushing latency (0.46s, ~4× faster than Whisper, ~28× faster than IndicWhisper). But Whisper Medium captures the locality name in 60% of utterances against Deepgram's 55% — the production-relevant metric tilts the other way. Sarvam matches Whisper on Fuzzy Score but trails on EHR and WER.

Out-of-domain: FLEURS Hindi, clean vs noisy

On FLEURS all models post WER near 1.0. This is largely a methodology artifact — FLEURS ground truth is Devanagari and the ITRANS round-trip introduces systematic token mismatch — so I read the clean-vs-noisy delta, not the absolute value. The delta produced a surprise: **Deepgram is the most affected by added Gaussian noise (+0.030 WER), Sarvam the least (+0.004).** Whisper and IndicWhisper sit between (+0.006 and +0.026). The likely read: Deepgram's smart-format pipeline is tuned for telephony-band noise, not the broadband Gaussian noise I synthesized; Sarvam appears more robust to that specific perturbation. Worth treating as directional on 30 samples, not conclusive.

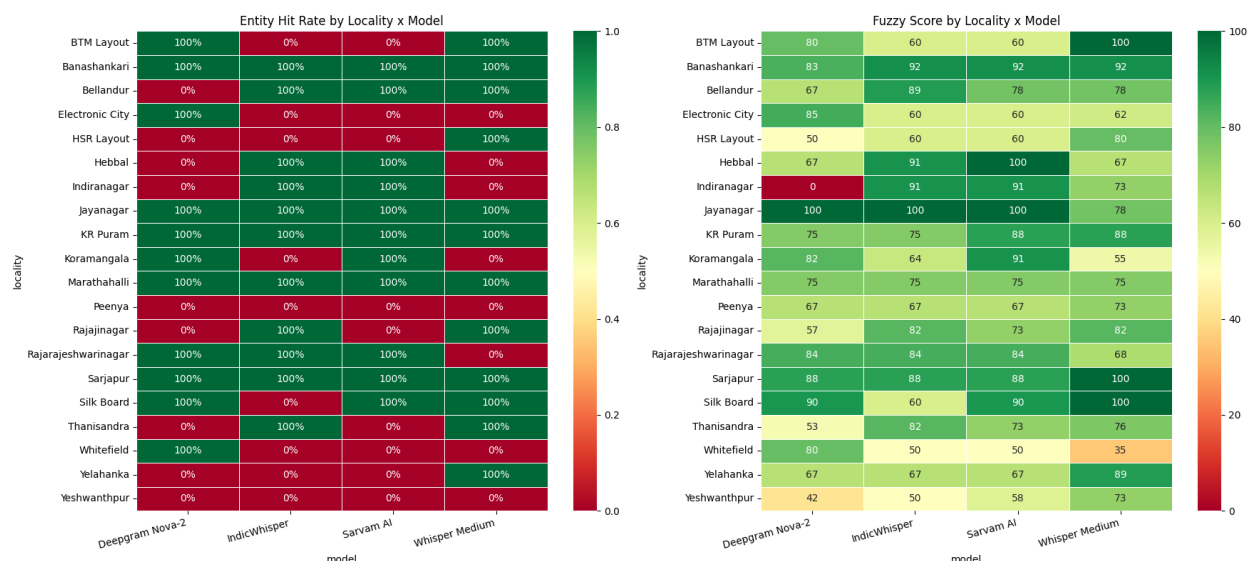


Failure Analysis

Failures cluster by **locality length and phonetic familiarity**, not by model. Common, shorter names (Jayanagar, Marathahalli, KR Puram, Sarjapur, Banashankari) hit 100% across the board. The hard cases are long, low-frequency Kannada names — and two of them defeat every model in the lineup:

- **Peenya and Yeshwanthpur — 0% EHR across all four models.** Yeshwanthpur is transcribed as “Jaisalmer-pur”, “jaisvant pur”, “jay vant pur”; Peenya is almost always dropped entirely. Every model fails the same way.
- **Whitefield, HSR Layout, Electronic City, Yelahanka — 25% EHR.** Whitefield is consistently rendered as “white field” / “vāya pheld” after Devanagari transliteration, which breaks the partial-ratio match even though the entity is clearly there. This is a normalization-pipeline failure as much as an ASR failure.

Two systemic patterns: (1) **the script boundary hurts more than the language boundary** — models often handle Hinglish carrier phrases fine but flip to Devanagari precisely on the proper noun, where the ITRANS round-trip then mangles it. (2) **Failures are correlated across models**, suggesting limits of training data (these names are rare in Hindi corpora) rather than model architecture.



Recommendation

Ship Deepgram Nova-2 for production, with a Whisper Medium fallback worth piloting. Deepgram wins on the constraints that matter most for telephony — 0.46s latency, lowest WER, hosted reliability. Whisper Medium’s 60% EHR vs Deepgram’s 55% is a real edge on the entity question, but its 1.64s latency and self-hosted operational burden mean it earns a pilot, not a default deployment.

- **Do not deploy IndicWhisper as-is.** 13.1s per utterance disqualifies it for real-time use, and the Hindi fine-tune did not buy enough entity-capture gain to justify the cost. Batch/offline only.
- **Sarvam AI** is worth a deeper look despite the highest WER — its Fuzzy Score (77.1) matches Whisper, and on FLEURS it was the most noise-robust of the four. Possibly better at entities than full sentences; the sample size here is too small to commit either way.
- **The model choice is not where the biggest win is.** Every model failed Peenya and Yeshwanthpur. Post-processing ASR output against a Bangalore locality gazetteer with fuzzy matching would lift EHR more cheaply than any model upgrade, and it works on top of whichever vendor wins.

Concrete next step: deploy Deepgram + a locality gazetteer post-processor. Run Whisper Medium in shadow mode on production traffic for two weeks; if its EHR edge holds at scale, revisit.

Limitations

- 20 self-recorded samples, single speaker — small and not demographically diverse. Directional, not statistically robust.

- WER is inflated by ITRANS normalization; absolute values are not comparable to published Hindi WER benchmarks.
- Sarvam free-tier rate limits (HTTP 429) required inter-request pacing; reported latency measures API response time only, with pacing overhead excluded.
- FLEURS clean/noisy split was 19/11 from the seeded Bernoulli draw; absolute WER is also affected by the Devanagari/Roman mismatch, so only the clean-vs-noisy delta is interpreted.